

Data Science Internship| Level 3 – Task 1

Objective :-

The objective of this task is to build regression models to predict the aggregate rating of restaurants using available features and evaluate their performance.

Code :-

```
196 from sklearn.model_selection import train_test_split
197 from sklearn.linear_model import LinearRegression
198 from sklearn.tree import DecisionTreeRegressor
199 from sklearn.ensemble import RandomForestRegressor
200 from sklearn.metrics import mean_absolute_error, mean_squared_error
201
202 # ----- Features & Target -----
203 X = df[['Average Cost for two', 'Price range', 'Votes',
204        | 'Table Booking', 'Online Delivery']]
205
206 y = df['Aggregate rating']
207
208 # ----- Train-Test Split -----
209 X_train, X_test, y_train, y_test = train_test_split(
210     X, y, test_size=0.2, random_state=42)
211
212 # ----- Models -----
213 models = {
214     "Linear Regression": LinearRegression(),
215     "Decision Tree": DecisionTreeRegressor(),
216     "Random Forest": RandomForestRegressor()
217 }
218
219 # ----- Training & Evaluation -----
220 print("\n--- Model Performance ---")
221
222 for name, model in models.items():
223     model.fit(X_train, y_train)
224     y_pred = model.predict(X_test)
225
226     mae = mean_absolute_error(y_test, y_pred)
227     mse = mean_squared_error(y_test, y_pred)
228
229     print(f"\n{name}:")
230     print("MAE:", mae)
231     print("MSE:", mse)
```

Output :-

```
--- Model Performance ---
```

```
Linear Regression:
```

```
MAE: 1.0750712721369122
```

```
MSE: 1.6764802747031424
```

```
Decision Tree:
```

```
MAE: 0.28511505226345507
```

```
MSE: 0.2024431846978273
```

```
Random Forest:
```

```
MAE: 0.23700978068945786
```

```
MSE: 0.13326202456085276
```

Model Building :-

The following features were used for prediction:

- Average Cost for Two
- Price Range
- Votes
- Table Booking
- Online Delivery

The dataset was split into training (80%) and testing (20%) sets.

Models Used :-

Three regression algorithms were implemented:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Model Evaluation :-

The models were evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

Results :-

- **Linear Regression**
 - **MAE: 1.075**
 - **MSE: 1.676**
- **Decision Tree**
 - **MAE: 0.285**
 - **MSE: 0.202**
- **Random Forest**
 - **MAE: 0.237**
 - **MSE: 0.133**

Analysis :-

- **Linear Regression performed poorly due to its inability to capture complex relationships in the data.**
- **Decision Tree showed significant improvement by modelling non-linear patterns.**
- **Random Forest performed the best, achieving the lowest error values due to its ensemble learning approach.**

Conclusion :-

Among all models, the Random Forest Regressor provided the best performance with the lowest prediction error. This indicates that ensemble methods are more effective for this dataset.

The model can be further improved by adding more relevant features and tuning hyperparameters.

Data Science Internship | Level 3 – Task 2

Objective :-

The objective of this task is to analyse the relationship between different cuisines and restaurant ratings, identify the most popular cuisines based on customer votes, and determine which cuisines tend to receive higher ratings.

Code :-

```
238 print("\n--- Cuisine Analysis ---")
239
240 # ✗ DO NOT split again
241 # df['Cuisines'] = df['Cuisines'].str.split(', ') ← remove this
242
243 # ✔ Only explode
244 cuisines_exploded = df.explode('Cuisines')
245
246 # ----- Average Rating -----
247 print("\n--- Average Rating by Cuisine ---")
248 cuisine_rating = cuisines_exploded.groupby('Cuisines')['Aggregate rating'].mean().sort_values(ascending=False)
249 print(cuisine_rating.head(10))
250
251 # ----- Votes -----
252 print("\n--- Most Popular Cuisines (Votes) ---")
253 cuisine_votes = cuisines_exploded.groupby('Cuisines')['Votes'].sum().sort_values(ascending=False)
254 print(cuisine_votes.head(10))
255
256 # ----- Count -----
257 print("\n--- Most Common Cuisines ---")
258 cuisine_count = cuisines_exploded['Cuisines'].value_counts()
259 print(cuisine_count.head(10))
260
```

Output :-

```
--- Cuisine Analysis ---

--- Average Rating by Cuisine ---
Cuisines
Sunda          4.900000
Börek          4.700000
Taiwanese      4.650000
Ramen          4.500000
Dim Sum        4.466667
Hawaiian       4.412500
Döner          4.400000
Bubble Tea     4.400000
Curry          4.400000
Kebab          4.380000
Name: Aggregate rating, dtype: float64

--- Most Popular Cuisines (Votes) ---
Cuisines
North Indian   595981
Chinese        364351
Italian        329265
Continental    288255
Fast Food      184058
American       183117
Cafe           177568
Mughlai        151946
Desserts       105889
Asian          104303
Name: Votes, dtype: int64

--- Most Common Cuisines ---
Cuisines
North Indian   3960
Chinese        2735
Fast Food      1986
Mughlai         995
Italian        764
Bakery         745
Continental    736
Cafe           703
Desserts       653
South Indian   636
Name: count, dtype: int64
```

Average Rating by Cuisine :-

The average rating was calculated for each cuisine.

Key Findings:

- Some cuisines have very high ratings such as:
 - Sunda (4.9)
 - Taiwanese (4.65)
 - Ramen (4.5)
 - Dim Sum (4.47)

👉 These cuisines are highly rated, indicating excellent customer satisfaction.

Most Popular Cuisines (Based on Votes) :-

The total number of votes was analysed to identify popular cuisines.

Top Cuisines:

- North Indian → Highest votes
- Chinese
- Italian
- Continental
- Fast Food

👉 These cuisines are the most preferred by customers.

Most Common Cuisines :-

The frequency of cuisines in the dataset was analysed.

Key Observations:

- North Indian is the most common cuisine
- Followed by Chinese, Fast Food, Mughlai, and Italian

👉 This shows these cuisines dominate the restaurant market.

Analysis :-

- Popular cuisines like North Indian and Chinese have high customer engagement (votes)
- However, highest-rated cuisines are not always the most common, indicating niche cuisines can perform exceptionally well
- Customer preference and quality do not always align perfectly

Conclusion :-

The analysis shows that while some cuisines are widely popular and commonly available, others achieve higher ratings despite being less common. This suggests that uniqueness and quality can significantly impact customer satisfaction.

Data Science Internship | Level 3 – Task 3

Objective :-

The objective of this task is to create visualizations to understand the distribution of restaurant ratings, compare ratings across cities and cuisines, and analyse relationships between different features and the target variable.

Code :-

```
266 import matplotlib.pyplot as plt
267 import seaborn as sns
268
269 # ----- PART 1: Distribution of Ratings -----
270 print("\n--- Rating Distribution ---")
271
272 plt.figure(figsize=(6,4))
273 plt.hist(df['Aggregate rating'], bins=20)
274 plt.title("Distribution of Ratings")
275 plt.xlabel("Rating")
276 plt.ylabel("Count")
277 plt.show()
278
279 # Bar plot (rating counts)
280 plt.figure(figsize=(6,4))
281 df['Aggregate rating'].value_counts().sort_index().plot(kind='bar')
282 plt.title("Rating Count")
283 plt.xlabel("Rating")
284 plt.ylabel("Number of Restaurants")
285 plt.show()
286
287
288 # ----- PART 2: Average Rating by City -----
289 print("\n--- Average Rating by City ---")
290
291 top_cities = df['City'].value_counts().head(10).index
292 city_rating = df[df['City'].isin(top_cities)].groupby('City')['Aggregate rating'].mean()
293
294 plt.figure(figsize=(8,5))
295 city_rating.sort_values().plot(kind='barh')
```



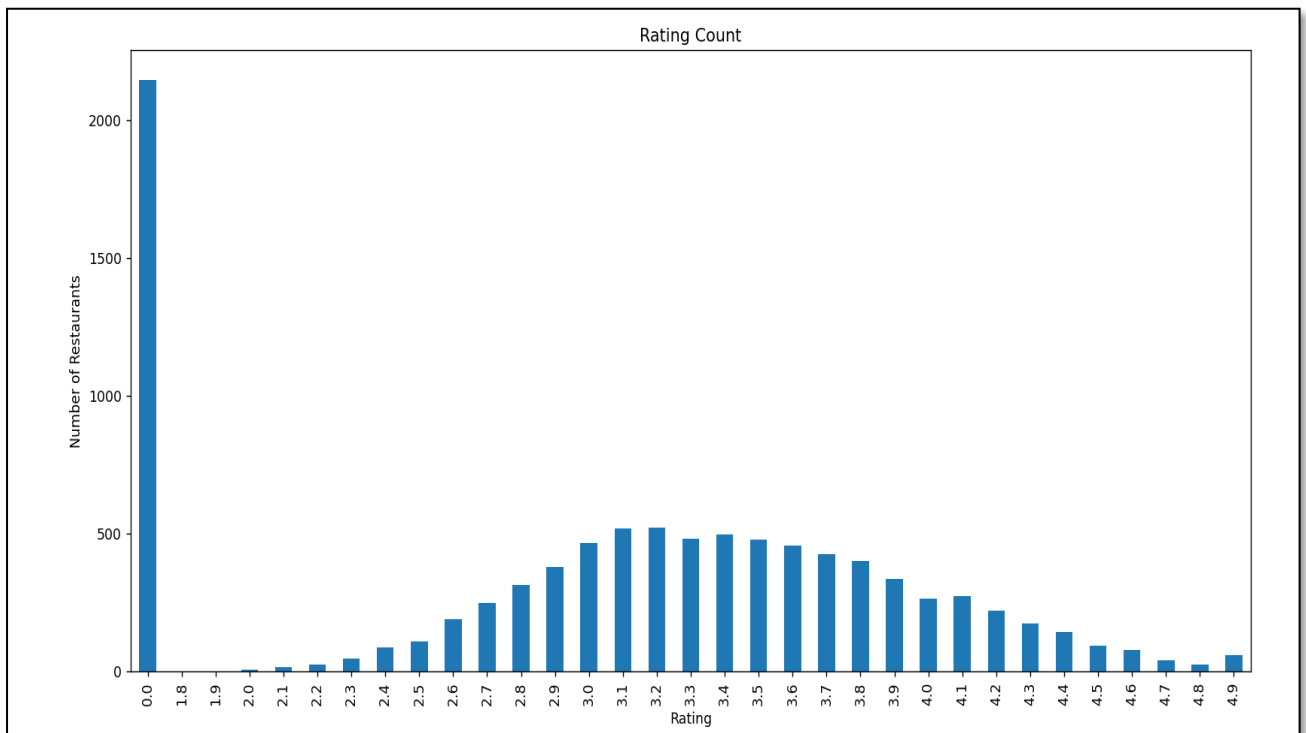
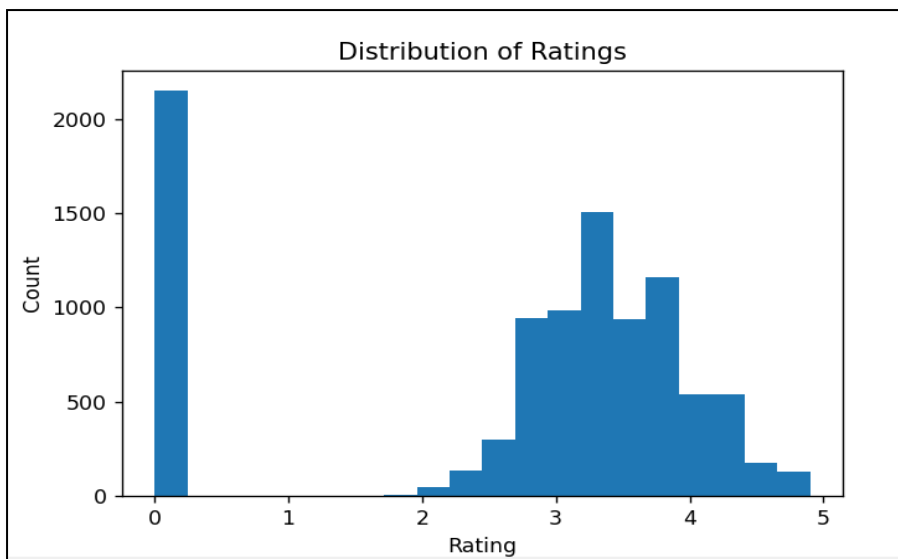
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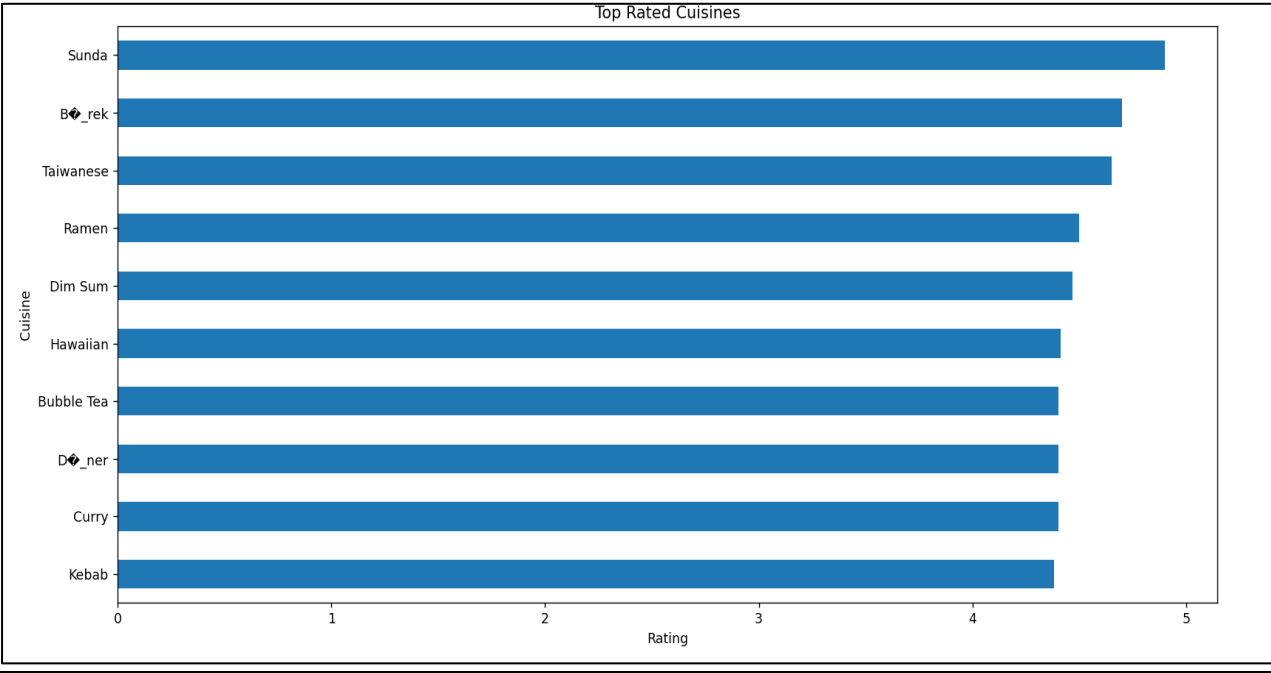
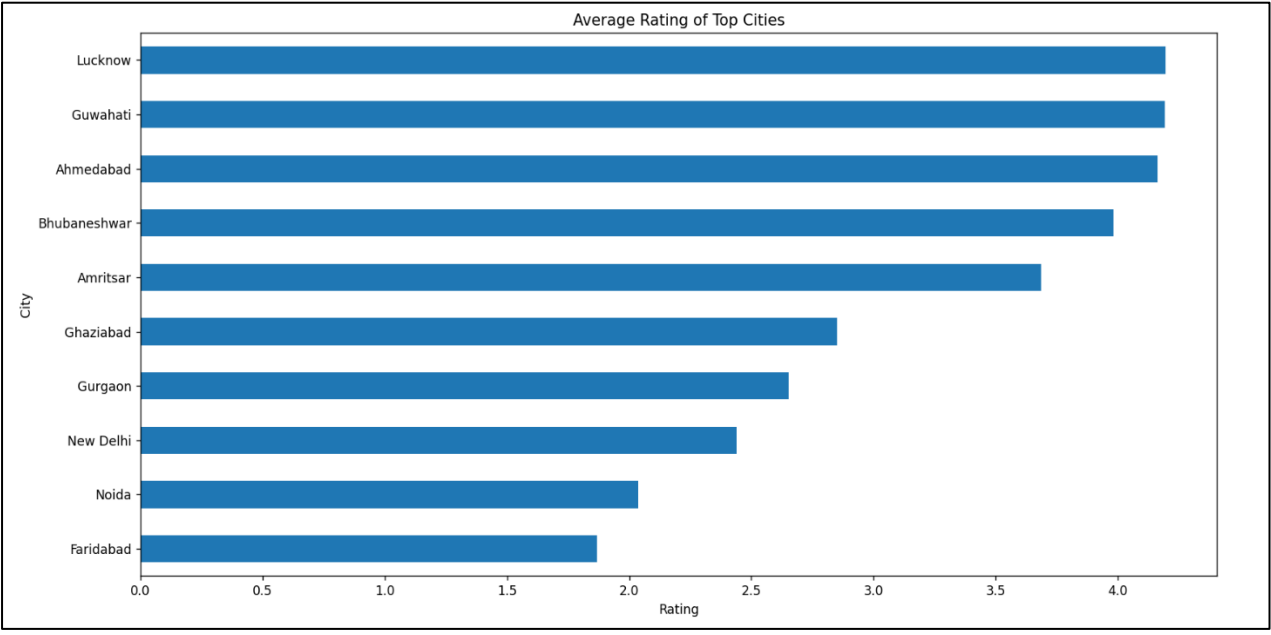
295 city_rating.sort_values().plot(kind='barh')
296 plt.title("Average Rating of Top Cities")
297 plt.xlabel("Rating")
298 plt.ylabel("City")
299 plt.show()
300
301
302 # ----- PART 3: Average Rating by Cuisine -----
303 print("\n--- Average Rating by Cuisine ---")
304
305 # Use exploded data from previous task
306 cuisine_rating = cuisines_exploded.groupby('Cuisines')['Aggregate rating'
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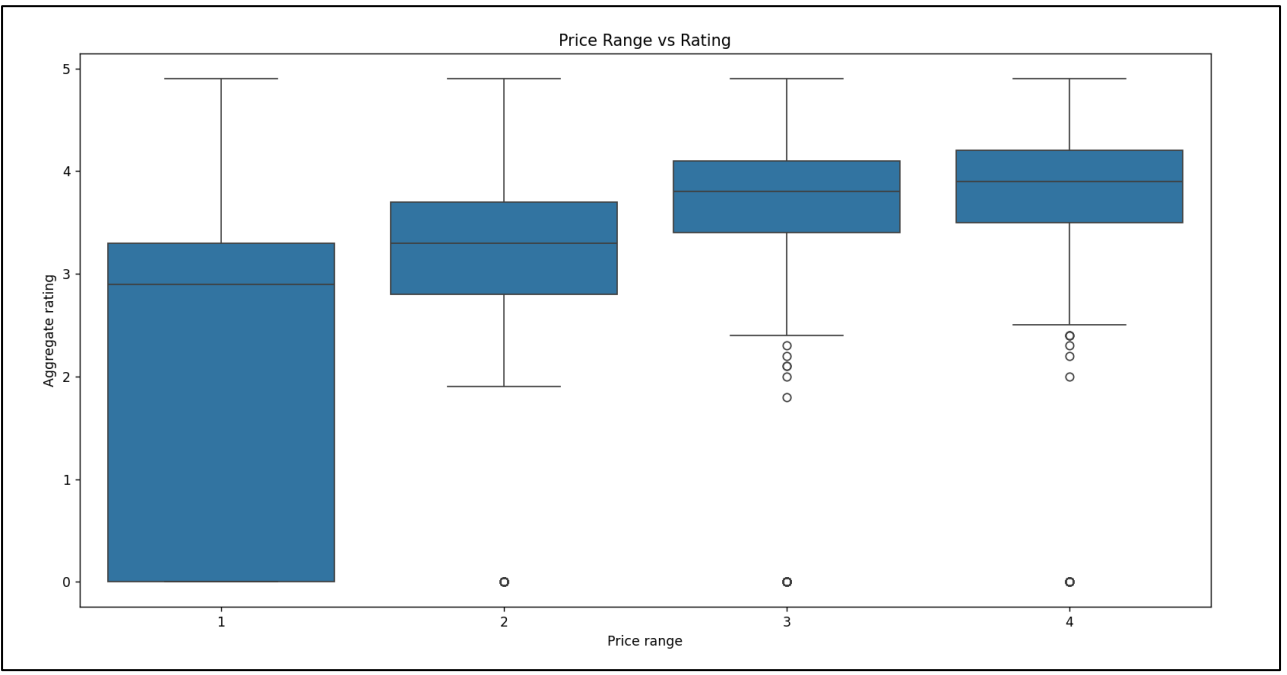
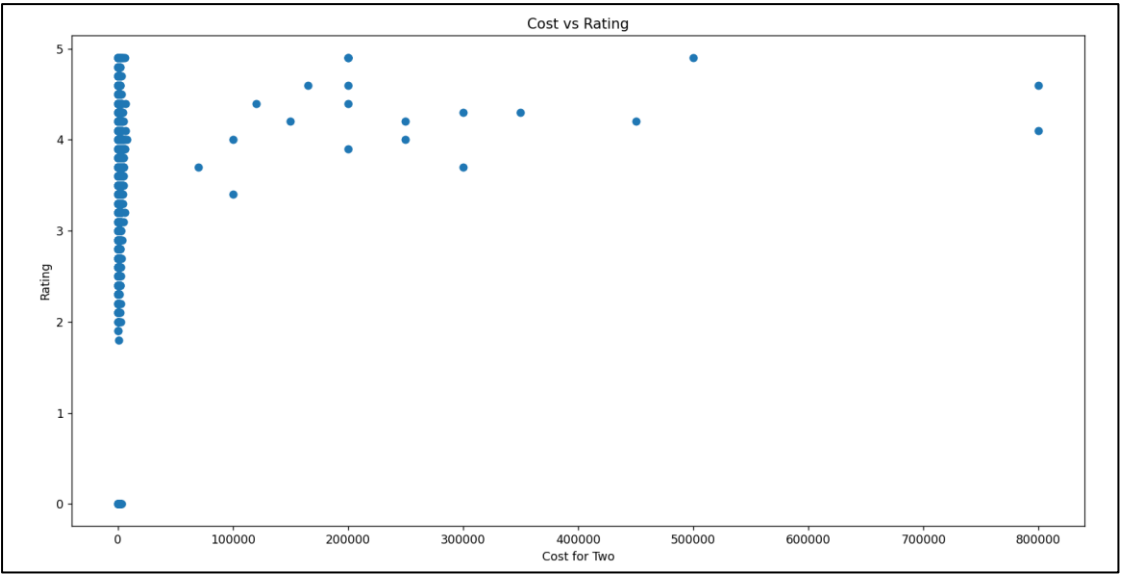
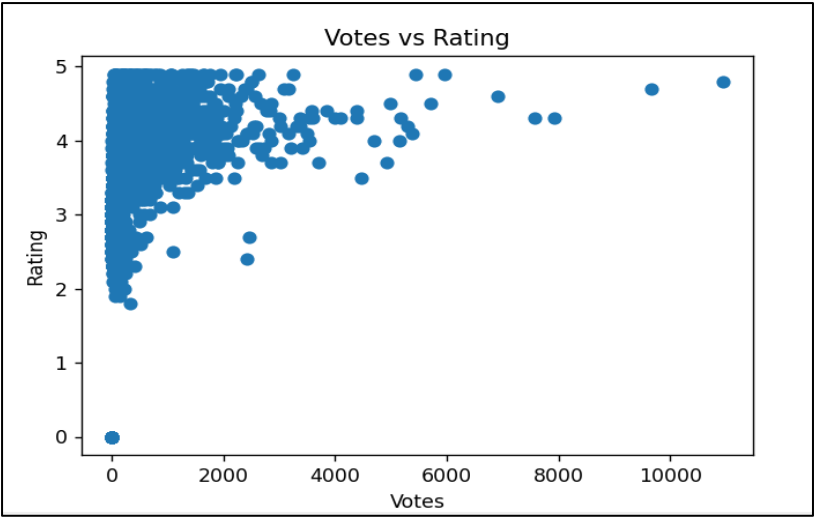
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Output :-

```
--- Rating Distribution ---  
  
--- Average Rating by City ---  
  
--- Average Rating by Cuisine ---  
  
--- Feature Relationships ---
```







Distribution of Ratings :-

Histograms and bar plots were used to visualize the distribution of ratings.

Observations:

- Most restaurant ratings fall between 2.0 and 4.0
- A significant number of restaurants have low ratings (0.0)
- Very high ratings (above 4.5) are relatively rare

👉 This indicates that the majority of restaurants have average to good ratings.

Ratings Across Cities and Cuisines :-



City-wise Analysis:

- Some cities show higher average ratings than others
- Ratings vary depending on location and customer base



Cuisine-wise Analysis:

- Certain cuisines have higher average ratings
- Niche cuisines tend to perform better in terms of quality

👉 This shows that both location and cuisine type influence ratings.

Relationship Between Features and Ratings :-

Key Insights:

- Votes vs Rating:
 - Restaurants with more votes generally have higher ratings
 - Indicates popularity and trust
- Cost vs Rating:
 - Slight positive trend
 - Higher cost restaurants often have better ratings
- Price Range vs Rating (Boxplot):
 - Higher price ranges tend to have higher median ratings

👉 This suggests that quality and customer engagement impact ratings significantly.

Conclusion :-

The visual analysis highlights that restaurant ratings are influenced by multiple factors such as popularity, price, cuisine, and location. Most restaurants fall in the mid-rating range, while higher ratings are associated with better service, higher cost, and strong customer engagement.